

Technical Reference Document: Esports Data Infrastructure

Document Purpose: This document establishes the technical architectures, cost models, and deployment timelines required to legally and securely utilize video game match telemetry for automated transaction settlement. It is designed to assist software engineers, product owners, and founders in evaluating third-party API integration strategies versus algorithmic computer vision workarounds.

1. The Core Engineering Challenge

Third-party platform developers cannot legally scrape public gaming APIs or community data endpoints to facilitate cash-prize match play without running into intellectual property boundaries. Major publishers (e.g., Epic Games, Riot Games) strictly restrict commercial monetization of their telemetry via standard developer licenses.

To resolve this conflict, a platform must either route its data requests through licensed commercial data networks (aggregators) that hold global B2B distribution rights, or implement an out-of-game verification pipeline.

2. Commercial Data Aggregator Pricing Models

Enterprise data providers (such as GRID Data or PandaScore) pay extensive licensing fees directly to publishers for server-level access. They pass these costs to developers using a tiered subscription structure:

Tier 1: Open Access Programs (Sandbox Environment)

- **Cost:** \$0 per month (Free)
- **Target State:** Conceptual prototyping, database schema mapping, and testing webhook reception.
- **Operational Scope:** Restricts developers to non-commercial, pre-revenue usage. It provides real-time data access for specific titles (such as *Counter-Strike 2* and *Dota 2*) to ensure backend automation works perfectly before capital is spent.

Tier 2: Historical & Post-Match APIs (Production Scale-Up)

- **Cost:** From €400 to €1,000 per month (Approximately \$430 to \$1,100 USD), scaled per game title.
- **Target State:** Platforms utilizing asynchronous settlement loops.
- **Operational Scope:** Provides detailed, finalized game stats (e.g., final kill counts, headshot percentages, match winners) once a game server closes the match file.
- **Limitation:** Standard commercial tiers explicitly prohibit wagering-adjacent operations. Moving into cash prize structures will trigger automated compliance audits, pushing the platform into the Enterprise Tier.

Tier 3: Live Pro Enterprise Feeds (Commercial Scale)

- **Cost:** \$2,000 to \$5,000+ per month (Subject to custom B2B evaluation and annual commitments).
- **Target State:** Scaled platforms requiring low-latency, millisecond-by-millisecond streaming WebSockets.

- **Operational Scope:** Bulletproof legal and technical coverage. It offers authorized real-time event feeds directly linked to server state, backed by compliance structures that allow for direct, real-money prize distribution.

3. The No-API Alternative: Computer Vision Pipeline

For early-stage startups needing a low-overhead, legally compliant minimum viable product (MVP), platforms can completely bypass game developer APIs through an **Asynchronous OCR (Optical Character Recognition) Pipeline**.

The Architectural Workflow:

1. **Image Ingestion:** Upon match completion, users take an unedited, high-resolution screenshot or video clip of the finalized in-game scoreboard and upload it to the application frontend.
2. **OCR Processing Layer:** The backend automatically routes the image asset to a cloud-based computer vision model (e.g., Google Cloud Vision API or AWS Rekognition).
3. **Data Extraction & Sanitization:** The vision model isolates and extracts specific text bounding boxes containing metrics like **Kills**, **Deaths**, **Accuracy**, or **Match Result**.
4. **Automated Settlement:** The system matches the string data against the challenge parameter rules and securely executes database adjustments to pay the winner out of escrow.

Strategic Impact:

This methodology drops fixed data-infrastructure overhead from thousands of dollars per month down to a variable fee of roughly \$0.0015 per image processing call. Furthermore, it completely isolates the application backend from third-party developer API keys, removing the risk of sudden developer-side bans.

4. Recommended Infrastructure Rollout Timeline

A phased engineering strategy allows a platform to validate its code architecture, user adoption metrics, and monetization streams before investing heavy capital into enterprise contracts.

Phase 1: Sandboxed Prototyping (Months 1 – 3)

- **Strategic Goal:** Validate database architecture, transaction tables, and wallet services.
- **Data Source:** Free Aggregator Open Access Tiers.
- **Fixed Infrastructure Cost:** \$0
- **Focus Area:** Developing internal data-parsing engines to transform game objects into structured database entries.

Phase 2: Commercial MVP Launch (Months 4 – 12)

- **Strategic Goal:** Establish real-money beta validation, prove user retention metrics, and build operational cash flows via processing rakes under a corporate structure (such as a Massachusetts LLC).
- **Data Source:** Asynchronous Computer Vision Pipeline / Manual User Verification.
- **Fixed Infrastructure Cost:** Variable pay-per-use OCR fees (fractions of a cent per match).

- **Focus Area:** Scaling fraud protection, managing manual dispute resolution queues, and establishing strict account age criteria to block cheating/smurfing.

Phase 3: Automated Enterprise Integration (Month 12+)

- **Strategic Goal:** Remove user upload friction, eliminate manual dispute overhead, and provide an instantaneous, automated user match experience.
- **Data Source:** Enterprise Real-Time Commercial Feeds (WebSockets).
- **Fixed Infrastructure Cost:** \$2,000 to \$5,000+ per month.
- **Focus Area:** Fully migrating match verification back to direct server-to-server endpoints, funded entirely by steady platform transaction rakes.